



Explain Concept Review in details with sub steps

Concept Review is the checkpoint where you evaluate different solution concepts against the requirements and decide which one deserves serious investment. Below is a detailed breakdown into practical substeps you can turn into review items.

1. Clarify goals and success criteria

- Define the specific questions this Concept Review must answer: technical feasibility, requirement coverage, business fit, risk level, etc.^{[1] [2]}
- Agree on go/no-go criteria up front (for example: must meet key requirements, acceptable risk, fits strategy and budget) so the team knows what “passing” means.^{[3] [1]}

2. Prepare concept documentation

- Create concise concept packages for each option: problem statement, target users, high-level architecture/mechanism, key features, basic sizing/capacity assumptions.^{[2] [1]}
- Add supporting inputs like customer pain points, early market research, competitor comparison, and rough cost/benefit expectations.^{[4] [3]}

3. Identify and invite stakeholders

- Select reviewers from engineering, design, manufacturing, testing, quality, marketing, service, and business/management so all perspectives are covered.^{[5] [1]}
- Clearly communicate their role: challenge assumptions, assess risks, and decide which concept(s) move on to preliminary design.^{[1] [2]}

4. Present concepts side by side

- Present each concept at a comparable level of detail (same template, same key views: function diagram, rough layout, main components, user interaction).^{[3] [4]}
- Highlight the main differences between concepts: working principle, complexity, expected performance, manufacturability, and novelty.^{[6] [4]}

5. Evaluate desirability, feasibility, viability

For each concept, systematically rate three dimensions (often called DFV):

- Desirability: how well it solves user problems and addresses market needs, including differentiation vs. competitors.^{[7] [4]}
- Feasibility: technical risk, availability of technology, internal know-how, expected development effort, and technical uncertainties.^{[5] [1]}
- Viability: business impact (margin, revenue potential), alignment with company strategy, and expected lifecycle cost.^{[4] [3]}

(You can implement this in your app as scoring sliders or a 1–5 rating per dimension.)

6. Analyze risks and constraints

- Identify major technical, schedule, cost, and regulatory risks for each concept, not in detail but at a high level (e.g. “new material with unproven fatigue data”).^{[6] [5]}
- Capture dependencies: special suppliers, new processes, tooling investments, or IP/licensing constraints that might block later phases.^{[8] [5]}

7. Capture structured feedback

- During the review meeting, facilitate discussion and record feedback per concept: strengths, weaknesses, open questions, and requested investigations.^{[2] [1]}
- Use simple tools (scorecards, ranking sheets, or Delphi-style expert scoring) so you can aggregate opinions instead of relying on loudest voices.^{[4] [5]}

8. Compare and prioritize concepts

- Aggregate scores or qualitative assessments to rank concepts by overall promise; eliminate those with low value or unacceptable risk.^{[3] [4]}
- If needed, decide on a “primary” concept plus a backup/variant to carry forward for limited additional exploration.^{[9] [3]}

9. Decide and define next steps

- Make a clear decision: proceed, refine and re-review, combine concepts, or stop the project/idea.^{[1] [2]}
- For the approved concept, define concrete next actions for Preliminary Design: required analyses, experiments, supplier talks, and updated business case.^{[8] [1]}

10. Document outcomes and update roadmap

- Record the decision, rationale, selected concept description, key assumptions, and known risks; link this record to your requirements baseline.^{[2] [3]}
- Update your project or product roadmap to reflect the chosen concept and communicate the outcome to all stakeholders who did not attend the review.^{[10] [3]}

If you want, I can now help you convert these 10 substeps into a compact checklist format (items and yes/no questions) tailored for mechanical design reviews in your app.



1. <https://productschool.com/resources/glossary/concept-review>
2. <https://amoeboids.com/product-management/glossary/concept-review/>
3. <https://www.productplan.com/glossary/concept-review>
4. <https://www.nextapp.co/blog/what-is-a-concept-review-a-tool-for-making-product-decisions>
5. https://craft.io/crf_glossary/what-is-a-conceptreview/
6. [https://www.emerald.com/jedt/article/18/5/1139/435374/Concept-evaluation-in-new-product-development A-set](https://www.emerald.com/jedt/article/18/5/1139/435374/Concept-evaluation-in-new-product-development-A-set)
7. <https://maze.co/guides/concept-testing/product/>
8. <https://mpconsulting.com/product-concept-review/>
9. <https://www.parallelhq.com/blog/concept-development-testing>
10. <https://learningloop.io/glossary/concept-review>
11. https://www.designsociety.org/download-publication/32334/concept_exploration_in_new_product_development—an_empirical_study
12. <https://slickplan.com/blog/what-is-concept-development>
13. https://www.designsociety.org/download-publication/24073/implementation_of_the_phase_review_processes_in_the_new_product_development_a_successful_experience
14. <https://www.sciencedirect.com/science/article/abs/pii/S0957417408005897>
15. <https://www.scribd.com/document/23985227/Design-Review-Check-Sheets>